MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8621

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal

See figure 1

Mounting position: Any

Weight: 4 ounces (113.4 grams) nominal

ABSOLUTE RATINGS: Class AB1

Parameter:	F	Ef	Eb	Ec1	Ec2	Ehk	Ib
Unit:	MHz	Vac	Vdc	Vdc	Vdc	Vdc	mAdc
Maximum:	500	26.5±5%	2,000	-150	400	±150	250
TEST CONDITIONS:	---	26.5	1,000	Adj	300	0	150

ABSOLUTE RATINGS: Class AB1

Parameter:	Pg1	Pg2	Pp	T(seals & anode core)	tk	Cooling
Unit:	W	W	W	°C	sec (min)	Note 1
Maximum:	2	12	250	250	30	---
TEST CONDITIONS:	---	---	---	---	120	See note 2

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	—							
(B) 2212	Pulse emission	3	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ik = 1.5 a; prr = 11 ±1; tp = 4,500 µs(min); Ef = 23.8 Vac	0.65	II	Δik	---	200	ma
1301	Heater current	-		0.65	II	If	0.45	0.62	Aac
1261	Electrode voltage (grid)	-	Eb = 1,600 Vdc; Ec2 = 400 Vdc; Ec1/Ib = 100 mAdc	0.65	II	Ec1	-62.0	-76.0	Vdc
1256	Electrode current (screen)	-		0.65	II	Ic2	-7.0	+3.0	mAdc
1266	Total grid current	4	Eb = 2,000 Vdc; Ec1/Ib = 125 mAdc	0.65	II	Ic1	---	-15	µAdc
1266	Primary grid emission (control)	-	Ic1 = 70 mAdc; t = 120; anode and screen grid floating	0.65	II	Isg1	---	-250	µAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; t = 120; Ic2 = 100 mAdc; anode floating	0.65	II	Isg2	---	-250	µAdc
1372	Current division (long pulse, method A)	-	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; egk/ib = 1.0 a; prr = 11 ±1; tp = 4,500 µs (min)	0.65	II	egk ic1 ic2	6.0 --- ---	18.0 200 260	v ma ma
1366	Interelement leakage resistance (cold)	-	Supply voltage = 500 Vdc; Rs = 2.5 Meg; anode positive	0.65	II	Rg2p	200	---	Meg
	Quality conformance inspection, part 2								
1031	Low frequency vibration	-	No voltages applied	---	---	---	---	---	---
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	-	EIA standard shields No. 320 and 321, or equal	---	---	Cgp Cin Cout	--- 14.2 4.0	0.06 17.2 5.3	pF pF pF
1336	Heater-cathode leakage	-	Ehk = +150 Vdc Ehk = -150 Vdc	---	---	Ihk Ihk	--- ---	150 150	µAdc µAdc
---	Power output	5	Class AB1 amplifier; F = 1 MHz (min), 30 MHz (max); Eb = 1,600 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 100 mAdc; RL = 2,000 ohms; t = 300 (max); Pd = zero level as indicated by Ic1 = 10 µAdc (max)	---	---	Po	250	---	W (useful)
	Quality conformance inspection, part 3								
---	Life test	6	Stability; Group C; Ef = 27.8 Vac; Eb = 1,600 Vdc; Ec2 = 400 Vdc at t = 0 hrs, Ec1/Ib = 100 mAdc; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life-test end points:								
---	Stability	7		---	---	ΔIb	---	20	mAdc
1366	Interelement leakage resistance (cold)	-		---	---	Rg1g2	50	---	Meg
						Rg1k	50	---	Meg
						Rg2K	50	---	Meg
						Rg2p	50	---	Meg
---	Power output	-		---	---	Po	235	---	W (useful)
1143	Forced cooling	8, 9	Ec1/Ib = 250 mAdc	---	---	T(anode core	---	225	°C
1155	Coolant-pressure drop versus coolant flow (forced air)	8, 10	No voltages applied	---	---	---	---	0.35	In. H_2O
1011	Humidity	8		---	---	---	---	---	---
---	Humidity-test end point:								
1266	Total grid current	-		---	---	Ic1	---	-15	μAdc

NOTES:

1. At an anode dissipation of 250 watts and an incoming air temperature of 25°C maximum, a minimum airflow of 3.8 cubic feet per minute (cfm) at sea level shall pass through the anode cooler. At this flow of 3.8 cfm, the static pressure drop across the tube and socket shown on Drawing 246-JAN shall be approximately 0.30 inch of water. The pressure drop varies with the amount of escaping air and with the shape and construction of the air director. This rating applies at bias voltages less than 100 volts and frequencies less than 500 MHz. Air cooling on the tube base shall be increased with increasing negative grid bias or with increasing frequency, or a combination of both. In all cases of operation, a socket which provides forced-air cooling of the base shall be used and maximum seal temperature ratings shall not be exceeded. The airflow shall be applied before or simultaneously with electrode voltages, and may be removed simultaneously with them.

2. In all electrical tests involving heater voltage, the socket shown on Drawing 246-JAN, shall be used. Forced-air cooling is permitted at the rate of 4.0 cfm maximum for the base and anode, unless otherwise specified in the specific test conditions. A separate source may be used for the base anode, but neither shall exceed 4.0 cfm.

3. The input wave shape shall have a tr and a tf of 25 μs maximum each, and the slope of the top of the pulse shall be not greater than 0.5 percent with a ripple not to exceed 0.1 percent.

4. This test is to be the first test performed at the conclusion of the holding period.

5. Alternately the power output test may be performed as follows:

---	Power output	Class AB1 amplifier; F = 1 MHz (min), 30 MHz (max); Eb = 1,600 Vdc; Ec2 = 400 Vdc; Ec1/Ibo = 100 mAdc; Rℓ = 2,000 ohms; t = 300 (max); Pd/Po = 250 W (useful)	Ic1	---	10	μAdc

Drive power (Pd) may not be increased beyond the level which produces a grid current (Ic1) of 10 μAdc. For life-test end-point check of power output, Pd/Po = 235 W (useful).

6. Under the voltage conditions specified, the bias voltage shall be adjusted to produce the specified value of anode current during an initial adjustment period not to exceed 2 hours.

7. With the voltage conditions as specified for life test, stability, including the original value of Ec1 which was used to set Ib = 100 mAdc, the value of Ib shall be measured and it shall not have changed by more than 20 mAdc from the original value of 100 mAdc.

(B) 8. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

9. The force-cooling test shall be made as follows: At an ambient temperature of 25°C, both base and anode shall be cooled by applying an airflow of 3.8 cfm maximum at sea level from a single source using the infinite baffle system as shown on figure 2, or equal. At the test conditions specified herein, the anode core temperature shall not exceed the limits specified herein. Temperature shall be measured by means of a thermocouple located as follows: The thermocouple shall be embedded in the top of the core, adjacent to the cooler, by means of drilling a small hole, shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole to hold the thermocouple firmly in place. In all cases, good electrical continuity between the thermocouple and the metal area in close proximity shall be demonstrated before the cooling test can be performed.

10. An infinite baffle system, as shown on figure 2, or equal, with an airflow of 3.8 cfm at sea level shall be used. The static pressure drop shall be measured across the tube and socket.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

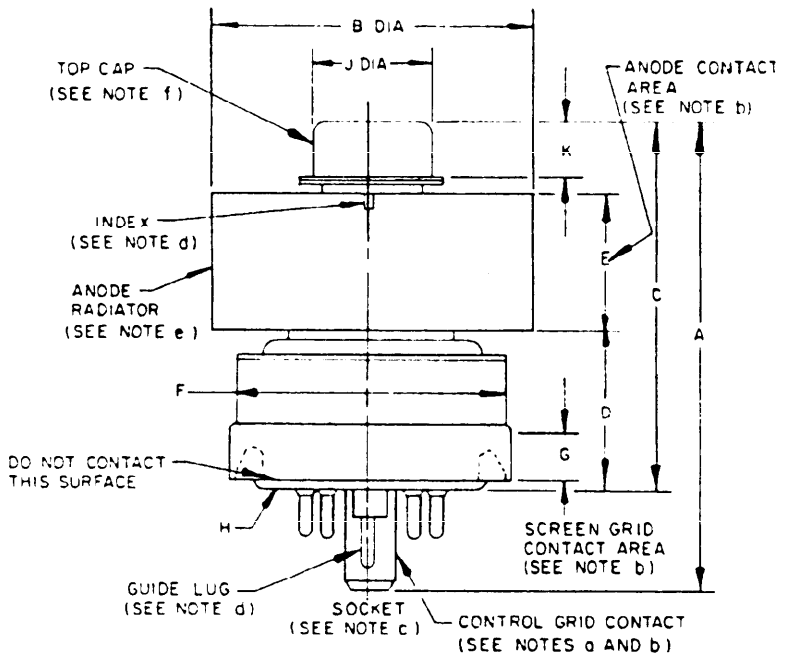
User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.324 (59.03)	2.464 (62.59)
C	1.810 (45.97)	1.910 (48.51)
Quality conformance inspection, part 3 (see note g)		
B	1.610 (40.89)	1.640 (41.66)
D	.750 (19.05)	.810 (20.57)
E	.710 (18.03)	.790 (20.07)
F		1.406 (35.71)
G	.187 (4.75)	
H	Base: B8-236 (see note a)	
J	.559 (14.20)	.573 (14.55)
K	.240 (6.10)	

PIN CONNECTIONS	
Pin No.	Element
1	g2
2	k
3	h
4	k
5	int con
6	k
7	h
8	k
center post	g1
radiator	a
base ring	g2

NOTES:

a. Pin alinement shall be checked by means of gage GB8-3. Dimensions of control-grid contact shall be inspected by means of gages specified on Drawing 246-JAN and shall be quality conformance inspection, part 2.

b. Alinement of anode, screen-grid, and control-grid contact surfaces shall be determined by means of gage specified on Drawing 168-JAN. Quality conformance inspection, part 2, shall apply.

c. Air-system socket shall be as specified on Drawing 246-JAN, EIMAC SK-600, or equal.

d. Location of guide lug of control-grid contact shall be referenced by a notch or arrow on the anode radiator in position shown.

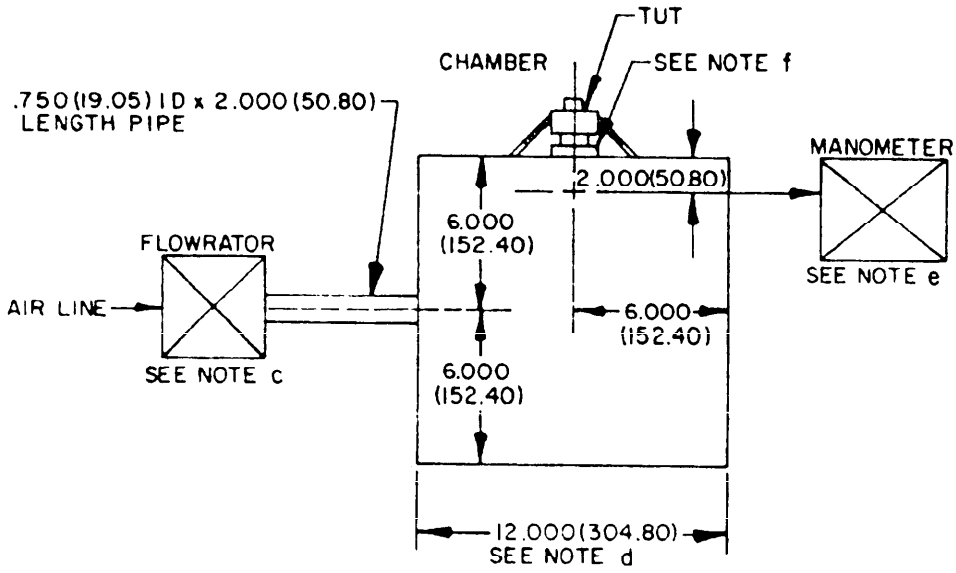
e. Anode clamping shall be confined to anode radiator.

f. Top cap outline optional provided it meets requirements of dimensions J and K.

Ⓑ g. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8621.

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NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parentheses.
c. Fisher Porter flowrator model B4-27-10/77, or equal.
d. 12 inch (304.80 mm) cube inside dimensions, compound sealed.
e. F.W. Dwyer manometer, 0 to 1 inch (25.40 mm) of water (Fisher Scientific Company 11-295-5 draft gage), or equal.
f. Socket specified on Drawing 246-JAN.

FIGURE 2. Baffle system.

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